UIC Department of
UNIVERSITY OF ILLINOIS AT CHICAGO Chemical Engineering
COLLEGE OF ENGINEERING

# assignment #1

**Course:** CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
**Term:** Spring 2020
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Your name: Majd Baradei
Due Date: Wednesday January 22 by 5:00pm in my office

Number of points............................100
Your score.................................. 90

**Note:** All problems are worth 100 points. You final score will be the average value

100/100

1a) Proving: $\left(\frac{\partial Cv}{\partial V}\right)_T = T\left(\frac{\partial^2 P}{\partial T^2}\right)_V$

Solution: We start with U(T,V) → dU = $\left(\frac{\partial U}{\partial T}\right)_V$ dT + $\left(\frac{\partial U}{\partial V}\right)_T$ dV

Since $\left(\frac{\partial U}{\partial T}\right)_V$ = Cv → dU = Cv dT + $\left(\frac{\partial U}{\partial V}\right)_T$ dV (1)

We know that dU = TdS – PdV

→ $\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P\left(\frac{\partial V}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P$ OK (2)

We know that A = U -TS → dA = dU - TdS – SdT ;However, dU = TdS - PdV

dA = - PdV – SdT

From A(V,T), we find the total differential dA:

$$dA = \left(\frac{\partial A}{\partial V}\right)_T dV + \left(\frac{\partial A}{\partial T}\right)_V dT$$

→ -P = $\left(\frac{\partial A}{\partial V}\right)_T$ and -S = $\left(\frac{\partial A}{\partial T}\right)_V$

Taking the the second derivative of dA we find that:

$$\frac{\partial}{\partial T}\left(\left(\frac{\partial A}{\partial V}\right)_T\right)_V = \frac{\partial}{\partial V}\left(\left(\frac{\partial A}{\partial T}\right)_V\right)_T \rightarrow \frac{\partial}{\partial T}(-P)_V = \frac{\partial}{\partial V}(-S)_T$$

→ $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$ we substitute in (2):

$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$ we substitute in (1):

dU = Cv dT + (T $\left(\frac{\partial P}{\partial T}\right)_V$ - P ) dV ; however, taking the second derivative of dU we find:

$$\frac{\partial}{\partial V}(Cv)_T = \frac{\partial}{\partial T}\left(\left(T\frac{\partial P}{\partial T}\right)_V - P\right)_V = \left(\frac{\partial T}{\partial T}\right)_V \cdot \left(\frac{\partial P}{\partial T}\right)_V + T\left(\frac{\partial^2 P}{\partial T^2}\right)_V - \left(\frac{\partial P}{\partial T}\right)_V$$

$$\rightarrow \frac{\partial}{\partial V}(Cv)_T = \left(\frac{\partial P}{\partial T}\right)_V + T\left(\frac{\partial^2 P}{\partial T^2}\right)_V - \left(\frac{\partial P}{\partial T}\right)_V$$

$$\rightarrow \left(\frac{\partial Cv}{\partial V}\right)_T = T\left(\frac{\partial^2 P}{\partial T^2}\right)_V$$

Very good!

86/100

1b) dS = $C_p \frac{dT}{T}$ -R $\frac{dP}{P}$ ; $C_p = A + B \frac{(\frac{C}{T})^2}{(\sinh(\frac{C}{T}))^2} + D \frac{(\frac{E}{T})^2}{(\cosh(\frac{E}{T}))^2}$

Since the expansion is isentropic → dS = 0

$\rightarrow 0 = C_p \frac{dT}{T} - R \frac{dP}{P} \rightarrow C_p = \frac{R\,T\,dP}{P\,dT}$

$$\rightarrow \frac{R\,T\,dP}{P\,dT} = A + B \frac{(\frac{C}{T})^2}{(\sinh(\frac{C}{T}))^2} + D \frac{(\frac{E}{T})^2}{(\cosh(\frac{E}{T}))^2} \qquad (x \frac{dT}{T})$$

$$\rightarrow \int \frac{R\,dP}{P} = A \int \frac{dT}{T} + B \int \frac{(\frac{C}{T})^2}{(\sinh(\frac{C}{T}))^2} \frac{dT}{T} + D \int \frac{(\frac{E}{T})^2}{(\cosh(\frac{E}{T}))^2} \frac{dT}{T}$$ We integrate both sides:

$$R \ln(\frac{P2}{P1}) = A\ln(\frac{T2}{T1}) + B \left[\frac{C}{T}\coth(\frac{C}{T}) - \ln(\sinh(\frac{C}{T}))\right]_{T1}^{T2} + D \left[\frac{-E}{T}\tanh(\frac{E}{T}) + \ln(\cosh(\frac{E}{T}))\right]_{T1}^{T2}$$

(-2)

$$R\ln(\frac{P2}{P1}) = R\ln\left[\frac{10^5 \frac{N}{m^2}}{30 \cdot 10^5 \frac{N}{m^2}}\right] = R\ln(\frac{1}{30}) \rightarrow$$

$F(T_2) = 0 \rightarrow$

$$A\ln(\frac{T2}{T1}) + B\left[\frac{C}{T_2}\coth(\frac{C}{T_2}) - \ln(\sinh(\frac{C}{T_2}))\right] - B\left[\frac{C}{T_1}\coth(\frac{C}{T_1}) - \ln(\sinh(\frac{C}{T_1}))\right] + D\left[\frac{-E}{T_2}\tanh(\frac{E}{T_2}) + \ln(\cosh(\frac{E}{T_2}))\right] - D\left[\frac{-E}{T_1}\tanh(\frac{E}{T_1}) + \ln(\cosh(\frac{E}{T_1}))\right] - R\ln(\frac{1}{30}) = 0$$

value of R = ? (-2)

We solve the following equation using the secant method: how? (-10)

| | | $T_2^{(0)}$ (C) | $T_2^{(0)}$ (K) | $T_2^{(1)}$ (K) | $F(T_2^{(0)})$ |
|---|---|---|---|---|---|
| A | 7.953090666 | | | | |
| B | 19.09166906 | -100 | 173.15 | 153.15 | 3.072609601 |
| C | 2086.9 | -120 | 153.15 | 110.2605135 | 2.095463891 |
| D | 9.936466991 | -162.8894865 | 110.2605135 | 118.7617129 | -0.518023956 |
| E | 991.96 | -154.3882871 | 118.7617129 | 117.7156723 | 0.072684451 |
| | | -155.4343277 | 117.7156723 | 117.681139 | 0.002322867 |
| $T_1$ (K) | 273.15 | -155.468861 | 117.681139 | 117.6812966 | -1.0645E-05 |
| $P_1$ (N/m²) | 30 | -155.4687034 | 117.6812966 | 117.6812965 | 1.55426E-09 |
| $P_2$ (N/m²) | 1 | -155.4687035 | 117.6812965 | 117.6812965 | 0 |
| | | -155.4687035 | 117.6812965 | #DIV/0! | 0 |
| | | | #DIV/0! | #DIV/0! | #DIV/0! |

We find that the secant method converges to 117.686 K $\rightarrow$ $T_2 = 117.686\text{K} - 273.15 = -155.464$ °C

**The temperature at the end of the expansion process $T_2 = -155.464$ °C**

85/100

1c) prove that PV = RT using $\frac{S}{R} = \ln[V \cdot (\frac{4\pi m U}{3h^2 N})^{\frac{3}{2}}] + \frac{5}{2}$ -5

→ $\frac{S}{R} = \ln(V) + \frac{3}{2}\ln[\frac{4\pi m}{3h^2 N}] + \frac{3}{2}\ln(U) + \frac{5}{2}$

We need to start from U = U (S,V) → dU = $(\frac{\partial U}{\partial S})_V$ dS + $(\frac{\partial U}{\partial V})_S$ dV

Therefor, we take the $(\frac{\partial U}{\partial S})_V$ of Sackur–Tetrode equation:

$\frac{1}{R} = 0 + 0 + \frac{3}{2}\frac{1}{U}(\frac{\partial U}{\partial S})_V$ → U = $\frac{3R}{2}(\frac{\partial U}{\partial S})_V$ how? (- 5) (1)

We take the $(\frac{\partial U}{\partial V})_S$ of Sackur–Tetrode equation: how? (- 5)

$0 = \frac{1}{V} + 0 + \frac{3}{2}\frac{1}{U}(\frac{\partial U}{\partial V})_S$ → U = $\frac{-3V}{2}(\frac{\partial U}{\partial V})_S$ (2)

From (1) and (2): $\frac{3R}{2}(\frac{\partial U}{\partial S})_V = \frac{-3V}{2}(\frac{\partial U}{\partial V})_S$ → $R(\frac{\partial U}{\partial S})_V = -V(\frac{\partial U}{\partial V})_S$ (3)

We also know that dU = T dS – P dV

→ $(\frac{\partial U}{\partial S})_V$ = T And $(\frac{\partial U}{\partial V})_S$ = -P ; we substitute it in (3):

RT = -V ( -P) → $\boxed{PV = RT}$